

Team Name	Ranking	small PLM	LLM	LSTM (+) CNN	fine-tuning	Data augmentation	CRF layer
TM-TREK _{STC:1}	1	✓			✓		✓
AIpom _{STC:2}	2		✓		✓		
USTC-BUPT _{STC:3}	3	✓	✓		✓	✓	
RKadiyala _{STC:6}	6	✓			✓		✓
DeepPavlov _{STC:7}	7	✓			✓	✓	
RUG-5 _{STC:17}	17	✓	✓		✓		
TueCICL _{STC:22}	22			✓			
jelarson _{STC:25}	25						
MasonTigers _{STC:27}	27	✓					
Unibuc-NLP _{STC:28}	28			✓	✓		

Table 8: **Subtask C** Participants method overview.

(Darwinkel et al., 2024).

5.4 Boundary Identification

Table 8 presents an overview of the methods used by the top-ranking systems for Subtask C. The best performing solutions are mainly based on ensemble strategies, with some employing data augmentation.

Team TM-TREK_{STC:1} (Qu and Meng, 2024) achieved the highest performance in Subtask C. They utilized an ensemble of XLNet models, each trained with a distinct seed, and used a straightforward voting mechanism on the output logits. They also explored the integration of LSTM and CRF layers on top of various PLMs, along with continued pretraining and fine-tuning of PLMs, and dice loss functions to enhance model performance.

Team AIpom_{STC:2} (Shirnin et al., 2024) introduced a novel two-stage pipeline merging outputs from an instruction-tuned, decoder-only (Mistral-7B-OpenOrca) model and two encoder-only sequence taggers.

Team USTC-BUPT_{STC:3} (Guo et al., 2024) fine-tuned a DeBERTa model with data augmentation and framed the task as a token classification problem.

Team RKadiyala_{STC:6} (Kadiyala, 2024) fine-tuned various encoder-based models with a Conditional Random Field (CRF) layer and found DeBERTa-V3 to perform the best on the development set.

Team DeepPavlov_{STC:7} (Voznyuk and Konovalov, 2024) fine-tuned the DeBERTa-v3 model using the provided dataset and developed a data prepro-

cessing pipeline for data augmentation.

Other teams explored diverse CNN, LSTM (Team TueCICL_{STC:22}) (Stuhlinger and Winkler, 2024), (Team Unibuc - NLP_{STC:28}) (Marchitan et al., 2024), and regression-based (Team jelarson_{STC:25}) (Larson and Tyers, 2024) techniques to address this challenge, although many did not surpass the baselines due to issues related to model overfitting or inadequate word embeddings.

6 Results and Discussion

6.1 Subtask A

There were three submissions for subtask A, which were submitted in time, but had the wrong file name, which prevented us from scoring them automatically. We eventually manually fixed the names and scored them, and we also added them to the ranking but marked them with a *. They should be considered as unofficial submissions.

Monolingual Table 9 presents the performance of systems in the monolingual track of Subtask A. Out of 125 participating teams, 15 surpassed the baseline, with the top-performing team (Genaïos) achieving an accuracy of 96.88. Notably, several teams demonstrated high precision and recall scores, indicating robust performance in distinguishing between human-generated and machine-generated text in a binary classification context.

Multilingual Table 10 presents the performance of systems in the multilingual track of Subtask A, where Team USTC-BUPT emerges as the top performer among 62 participating teams, achieving an accuracy of 95.99, remarkably close to the English-only result. Their methodology entails a blend of language detection and fine-tuning tasks using LLaMA-2-70B for English and the mT5 model for others, showcasing their adaptability across diverse languages.

Similarly, among the 22 teams surpassing the baseline, the majority leverage advanced LLMs such as LLaMA, Mistral, etc., while also emphasizing syntax and writing style differences between human and AI-generated texts. For example, Team FI Group implements a hierarchical fusion strategy to adaptively fuse representations from different BERT layers, prioritizing syntax over semantics for improved classification accuracy. Likewise, Team KInIT employs an ensemble approach, combining fine-tuned LLMs with zero-shot statistical methods,

Rank	Team	Prec	Recall	F1-score	Acc	Rank	Team	Prec	Recall	F1-score	Acc
*	dianchi	96.21	99.19	97.68	97.53	62	Collectivized Semantics	68.21	99.39	80.90	75.35
1	Genaios	96.11	98.03	97.06	96.88	63	IUCL	68.13	98.33	80.49	74.96
2	USTC-BUPT	95.75	96.86	96.30	96.10	64	annedadaa	68.01	97.69	80.19	74.66
3	mail6djj	94.87	97.18	96.02	95.76	65	cmly99	67.92	97.96	80.22	74.62
4	howudoin	93.48	98.12	95.74	95.42	66	xiaoll	67.92	97.96	80.22	74.62
5	idontknow	94.57	95.42	94.99	94.72	67	SINAI	67.31	99.88	80.42	74.46
6	seven	90.12	98.31	94.04	93.46	68	yuwert777	68.78	92.96	79.06	74.14
7	zongxiong	93.54	93.82	93.68	93.35	69	yaoxy	68.78	92.96	79.06	74.14
8	mahsaamani	90.59	96.23	93.32	92.77	70	moniszc	67.25	97.66	79.65	73.79
9	bennben	91.49	95.05	93.24	92.76	71	MasonTigers	67.59	95.72	79.23	73.64
10	infinity2357	91.92	90.96	91.43	91.05	72	AT	67.30	96.59	79.33	73.56
11	AISPACE	84.76	99.92	91.72	90.52	73	DUT	66.27	99.92	79.69	73.24
12	petkaz	85.54	98.59	91.61	90.51	74	surbhi	69.38	87.40	77.35	73.12
13	moniszc1	86.96	95.68	91.11	90.20	75	thanet	69.47	86.69	77.13	73.00
14	moniszc3	86.96	95.68	91.11	90.20	76	Kathlalu	74.47	73.89	74.18	72.98
15	flash	82.39	99.77	90.25	88.68	77	KInIT	66.14	98.44	79.12	72.71
*	baseline	93.36	84.02	88.44	88.47	78	iimasNLP	67.81	87.08	76.25	71.50
16	ericmxf	81.71	99.98	89.93	88.24	79	wwzzhh	64.38	99.49	78.18	70.82
17	HU	82.63	97.24	89.34	87.81	80	bharathsk	64.48	98.69	78.00	70.76
18	jrutkowski2	84.58	93.44	88.79	87.61	81	apillay2	64.48	98.69	78.00	70.76
19	lihaoran	89.26	85.86	87.53	87.15	82	ashinee20	71.91	71.57	71.74	70.39
20	TrustAI	89.21	85.50	87.31	86.95	83	longfarmer	63.81	99.03	77.61	69.99
21	TM-TREK	79.47	99.99	88.56	86.43	84	mlnick	63.61	100	77.76	69.96
22	jojoc	86.30	87.76	87.02	86.25	85	vasko	63.47	99.93	77.63	69.75
23	RFBS	91.58	80.64	85.76	85.95	86	Groningen F	72.74	65.62	68.99	69.02
24	L3i++	81.41	94.66	87.53	85.84	87	hhy123	62.76	99.94	77.11	68.83
25	art-nat-HHU	86.29	86.04	86.17	85.49	88	1024m	62.54	99.98	76.95	68.53
26	FI Group	79.52	96.99	87.39	85.30	89	lhy123	62.54	99.96	76.94	68.53
27	phuhoang	87.72	83.65	85.64	85.26	90	thang	62.25	99.98	76.73	68.14
28	Unibuc-NLP	78.01	99.86	87.59	85.14	91	nikich28	61.90	99.26	76.25	67.52
29	sushvin	82.65	89.76	86.06	84.73	92	niceone	61.70	98.39	75.84	67.08
30	NewbieML	79.32	95.06	86.48	84.39	93	pmalea	60.73	97.87	74.95	65.64
31	QUST	76.88	99.91	86.89	84.17	94	mahaalblooki	60.85	96.23	74.56	65.51
32	MLab	83.17	85.83	84.48	83.44	95	bertsquad	60.32	98.94	74.95	65.27
33	ziweizheng	82.31	85.01	83.63	82.53	96	jjonczyk	60.25	99.31	75.00	65.23
34	Alpom	74.34	99.97	85.27	81.86	97	dkoterwa	60.12	99.98	75.09	65.16
35	lyaleo	79.38	87.82	83.39	81.62	98	lystsoval	92.30	35.31	51.07	64.48
36	yunhfang	75.26	95.31	84.10	81.08	99	sunilgundapu	59.22	99.31	74.20	63.72
37	sankalpbahad	78.22	88.08	82.86	80.86	100	RUG-D	59.19	99.35	74.18	63.68
38	aktsvigun	78.22	88.08	82.86	80.86	101	RUG-5	60.79	84.13	70.58	63.17
39	NootNoot	78.22	88.08	82.86	80.86	102	harshul24	54.55	57.14	55.81	62.00
40	Mast Kalandar	74.65	96.16	84.05	80.83	103	basavraj10	54.55	57.14	55.81	62.00
41	I2C-Huelva	73.92	98.01	84.28	80.79	104	samnlptaskab	58.01	99.93	73.41	61.97
42	priority497	73.31	99.69	84.49	80.78	105	partnlu	57.87	99.96	73.31	61.76
43	wjm123	73.31	99.69	84.49	80.78	106	teams2024	57.78	99.97	73.23	61.61
44	scalar	73.10	99.97	84.45	80.67	107	Rkadiyala	57.30	99.98	72.85	60.86
45	werkzeug	75.28	93.88	83.56	80.59	108	rtuora	57.18	99.89	72.73	60.66
46	blain	72.51	99.96	84.05	80.07	109	teamlanlp2	56.93	99.71	72.48	60.24
47	xxm981215	72.32	99.79	83.86	79.83	110	jakubbebac	56.89	99.88	72.49	60.18
48	moyanxinxu	72.32	99.79	83.86	79.83	111	dandread	55.19	99.87	71.92	59.04
49	jrutkowskikag1	73.02	97.54	83.52	79.78	112	pask1	55.97	99.80	71.72	58.67
50	NCL-UoR	75.10	90.84	82.22	79.37	113	skillissue	55.14	100	71.09	57.27
51	Sharif-MGTD	73.41	93.75	82.35	78.89	114	RUG-3	54.92	99.73	70.83	56.87
52	wgm123	71.16	99.94	83.13	78.69	115	Mashee	57.11	59.58	58.32	55.27
53	logiczmaksimka	70.79	99.29	82.65	78.11	116	TueCICL	55.37	69.61	61.68	54.57
54	somerandomjj	70.43	98.55	82.15	77.51	117	RUG-1	52.52	100	68.87	52.52
55	totylkokuba	70.43	98.55	82.15	77.51	118	novice8	52.24	68.68	59.34	50.57
56	lly123	69.25	99.95	81.81	76.66	119	ronghaopan	52.49	38.47	44.40	48.70
57	mimkag2	69.69	97.99	81.45	76.56	120	kamer	52.89	29.56	37.93	48.48
58	priyansk	69.28	99.36	81.64	76.53	121	helenpy	75.29	1.79	3.51	48.11
59	nampfiey1995	77.92	76.93	77.43	76.44	122	ascisel	7.14	0.01	0.01	47.44
60	xiangrunli	68.23	99.97	81.11	75.54	123	laida	40.18	19.28	26.06	42.53
61	roywang	68.23	99.97	81.11	75.54	124	nz28555	40.31	33.18	36.40	39.10
*	badrock	71.13	89.50	79.27	75.41						

Table 9: Subtask A monolingual Prec (precision), Recall, and F1-scores(%) with respect to MGT.

Rank	Team	Prec	Recall	F1-score	Acc
1	USTC-BUPT	94.93	97.53	96.21	95.99
2	FI Group	94.28	98.00	96.10	95.85
3	KInIT	92.95	97.86	95.34	95.00
4	priyansk	90.70	98.14	94.28	93.77
5	L3i++	92.47	94.00	93.23	92.87
6	QUST	90.45	90.98	90.71	90.27
7	xxm981215	90.45	90.98	90.71	90.27
8	NCL-UoR	81.42	95.41	87.86	86.23
9	AIpom	80.72	95.80	87.61	85.85
10	RFBES	85.43	85.27	85.35	84.71
11	blain	76.12	98.67	85.94	83.14
12	xiangrunli	75.20	99.67	85.73	82.66
13	wgm123	75.20	99.67	85.73	82.66
14	roywang	75.08	99.75	85.68	82.58
15	logiczmaksimka	74.34	99.33	85.04	81.74
16	zaratiana	74.75	96.68	84.31	81.21
17	thanet	76.18	92.56	83.58	81.00
*	baseline	73.45	99.30	84.44	80.89
18	cmv99	73.29	99.61	84.45	80.83
19	lly123	73.09	99.67	84.33	80.65
20	moyanxinxu	73.09	99.67	84.33	80.65
21	SINAI	72.51	99.91	84.04	80.17
22	Unibuc-NLP	71.82	99.79	83.52	79.43
23	annedadaa	72.16	98.57	83.32	79.39
24	1024m	71.03	99.91	83.03	78.66
25	sunilgundapu	71.04	98.86	82.67	78.35
26	hirak	70.79	99.66	82.78	78.34
27	bertsquad	70.45	99.12	82.36	77.82
28	Rkadiyala	69.99	99.94	82.33	77.59
*	dianchi	69.88	99.91	82.24	77.46
29	lyaleo	69.50	99.83	81.95	77.03
30	werkzeug	69.33	99.81	81.82	76.83
31	mlnick	69.25	99.81	81.77	76.75
32	RUG-5	69.90	96.78	81.17	76.55
33	DUTh	68.95	99.93	81.60	76.45
34	dandread	68.31	99.75	81.09	75.69
35	Genaios	68.30	99.73	81.07	75.67
36	vasko	67.99	98.68	80.50	75.03
37	thang	67.16	99.78	80.29	74.40
38	mahsaamani	68.53	93.21	78.98	74.09
39	RUG-D	65.03	99.55	78.67	71.79
40	omarnasr	64.80	99.21	78.40	71.43
41	lhy123	64.62	99.85	78.46	71.36
42	priority497	64.62	99.85	78.46	71.36
43	hhy123	64.47	99.84	78.35	71.17
44	wjm123	64.47	99.84	78.35	71.17
45	aktsvigun	62.83	99.36	76.98	68.96
46	sankalpbad	62.83	99.36	76.98	68.96
47	NootNoot	62.83	99.36	76.98	68.96
48	nampfiev1995	61.37	77.15	68.36	62.70
49	MasonTigers	56.77	100	72.42	60.21
50	RUG-1	51.33	97.39	67.23	51.22
51	novice8	51.95	84.56	64.36	51.08
52	scalar	52.04	80.17	63.11	51.04
53	mahaalblooki	48.96	51.24	50.08	50.55
54	Sharif-MGTD	51.42	67.13	58.23	50.53
55	TrustAI	51.15	62.04	56.07	50.06
56	sky2024just	51.96	26.37	34.99	48.79
57	laida	48.88	20.56	28.94	47.27

Table 10: **Subtask A multilingual** Prec (precision), Recall, and F1-scores(%) with respect to **MGT**.

resulting in a unique combination of techniques that effectively enhances classification accuracy.

Overall, these successful methodologies under-

Rank	Team	Prec	Recall	F1-score	Acc
1	AISPACE	91.81	90.85	90.84	90.85
2	Unibuc - NLP	88.69	86.96	87.03	86.96
3	USTC-BUPT	89.54	84.33	82.72	84.33
4	dianchi	86.45	83.48	83.62	83.48
5	NootNoot	86.68	83.12	83.15	83.12
6	L3i++	86.01	83.12	83.08	83.12
7	MLab	85.00	82.67	82.76	82.67
8	werkzeug	86.30	82.23	81.63	82.23
9	flash	88.29	82.23	79.46	82.23
10	juse7198	86.83	82.03	80.72	82.03
11	idontknow	88.44	80.94	77.47	80.94
12	TM-TREK	86.42	79.84	79.46	79.84
13	howudoin	80.02	79.68	79.79	79.68
14	TrustAI	83.80	79.19	79.07	79.19
15	I2C-Huelva	84.45	78.90	78.82	78.90
16	ericmxf	85.52	78.74	76.88	78.74
17	MGTD4ADL	83.78	76.96	74.46	76.96
18	scalar	81.90	76.26	76.00	76.26
19	ronghaopan	81.11	75.19	71.38	75.35
20	sunilgundapu	81.06	75.06	73.81	75.06
*	baseline	81.14	74.61	72.59	74.61
21	Collectivized Semantics	82.35	73.87	70.26	73.87
22	priyansk	78.06	73.36	67.05	73.36
23	logiczmaksimka	67.73	69.13	64.36	69.13
24	annedadaa	79.55	68.98	64.55	68.98
25	hhy123	65.94	67.77	63.01	67.77
26	xiangrunli	65.94	67.77	63.01	67.77
27	wjm123	65.94	67.77	63.01	67.77
28	lhy123	65.94	67.77	63.01	67.77
29	lly123	65.94	67.77	63.01	67.77
30	wgm123	65.94	67.77	63.01	67.77
31	moyanxinxu	65.94	67.77	63.01	67.77
32	priority497	65.94	67.77	63.01	67.77
33	thang	66.36	67.68	63.79	67.68
34	blain	63.15	67.23	62.35	67.23
35	xxm981215	65.77	67.21	62.41	67.21
36	QUST	65.77	67.21	62.41	67.21
37	mahaalblooki	63.72	66.27	61.82	66.27
38	MasonTigers	73.62	65.04	64.47	65.04
39	Rkadiyala	65.81	64.91	59.98	64.91
40	1024m	66.10	64.38	59.82	64.38
41	RUG-5	62.21	64.21	59.04	64.21
42	thanet	63.42	61.88	55.58	61.88
43	mlnick	66.84	61.79	57.53	61.79
44	RUG-D	66.39	61.54	53.82	61.54
45	Groningen F	60.10	60.84	57.90	60.84
46	NCL-UoR	69.03	60.15	58.05	60.15
47	mahsaamani	60.41	59.42	52.89	59.42
48	dandread	71.95	58.35	52.28	58.35
49	DUTh	63.71	56.68	51.25	56.68
50	bertsquad	57.27	55.97	51.49	55.97
51	RUG-3	61.51	54.23	49.26	54.23
52	cmv99	58.04	53.35	50.58	53.35
53	skysky12	60.86	53.31	50.14	53.31
54	vasko	59.98	52.82	50.38	52.82
55	phuhoang	61.54	50.79	50.21	50.79
56	rtuora	54.21	50.32	44.15	50.32
57	AT	53.13	48.59	43.91	48.59
58	teams2024	45.50	47.01	41.05	47.01
59	windwind22	39.87	39.31	32.79	39.31
60	helenpy	39.88	38.27	32.20	38.27
61	iimasNLP	39.88	38.27	32.20	38.27
62	clulab-UofA	37.53	29.29	24.58	29.29
63	samlptaskab	25.78	27.81	21.07	27.81
64	mhr2004	17.06	17.06	17.06	17.06
65	xiaoll	5.73	17.15	8.47	17.02
66	surbhi	17.24	16.77	15.10	16.77
67	roywang	2.78	16.67	4.76	16.67
68	RUG-1	2.78	16.67	4.76	16.67
69	novice8	16.39	16.55	13.93	16.55
70	NewbieML	15.99	15.58	14.13	15.30

Table 11: **Subtask B: Multi-Way Generator Detection** Prec (precision), Recall, and F1-scores(%) macro average.